Reading: the woman who tested the water

Nom _____

Grup _____

Data _____

Lectura per a casa després de la sessió 6. El debat es fa a la sessió 7. Llegeix el text, consulta el glossari i respon les preguntes.

Ellen Swallow Richards (1842–1911)



Ellen Swallow was born in 1842 in Massachusetts, in the United States. As a girl, she loved science. At that time, very few universities accepted women. Ellen studied at Vassar, a college for women. Then, in 1870, she asked to study chemistry at the Massachusetts Institute of Technology (MIT). MIT accepted her as “an experiment”. She was the first woman at MIT, and in 1873 she finished her degree in chemistry.

In the 1880s, many towns in Massachusetts were growing very fast. People drank water from rivers, lakes and wells, but some of this water was dirty. Waste water from houses and factories went into the rivers, and many people got ill.

In 1887, the government of Massachusetts asked Ellen and her team for help. For two years, they analysed more than 20,000 samples of water from all over the state. In the laboratory, they measured the amount of chloride in each sample. Chloride comes from salts, like sodium chloride, the salt we use in the kitchen. Clean water far from the sea has only a small, normal amount of chloride. When a sample had more chloride than normal, it was a sign of pollution from waste water.

Ellen made a map with the normal values for the whole state: the “Normal Chlorine Map”. Thanks to this work, Massachusetts created the first water-quality standards in the United States, and a city built one of the first modern plants to clean waste water.

Today, scientists still use Ellen's idea: to know if water is clean, we compare a sample with a reference.

Glossary: sample = mostra · waste water = aigües residuals · well = pou · to get ill = emmalaltir · amount = quantitat · pollution = contaminació · standard = norma, valor de referència · plant = planta (instal·lació) · to clean = netejar, depurar

A Reading comprehension

1 **True or false?** Correct the false sentences.

Sentence	T / F	Correction
Ellen studied at MIT before she studied at Vassar.		
In 1870, it was easy for women to study at university.		
Ellen and her team analysed more than 20,000 water samples.		
A high amount of chloride was a sign of pollution.		
Ellen's work helped to create the first water-quality standards in the USA.		

2 **Answer in English with short sentences.**

a. Why did many people get ill in Massachusetts in the 1880s?

b. What did Ellen measure in each water sample?

c. What is the “Normal Chlorine Map”?

B La ciència del text

3 **L'ió clorur.** D'on prové el clorur de l'aigua? Quin pot del nostre repte en conté? Com el vam detectar a la sessió 5?

4 **Mostra i referència.** Ellen comparava cada mostra amb un valor normal. Quan nosaltres hem fet el mateix en aquesta SdA? Posa'n un exemple.

5 **Aigua i salut.** Per què és important que hi hagi normes de qualitat de l'aigua de l'aixeta? Busca d'on prové l'aigua de l'aixeta de Barcelona i cita la font.

C Per al debat de la sessió 7

6 **Una científica poc coneguda.** El treball d'Ellen va millorar la salut de milions de persones, però avui poca gent en coneix el nom. Escriu dues possibles raons a partir del text i del context de l'època. Pistes: en aquella època, les dones podien votar o ser professores d'universitat? Els descobriments aplicats, com tenir aigua neta, es fan tan famosos com els grans descobriments teòrics?